

Volume II · Chapter 8 — Prompt Engineering for Medicine · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-8-context.md` for the AI interaction log.

Prerequisites: Ch 7.

Learning outcomes — the student can: design effective prompts (zero/few-shot, chain-of-thought); produce structured outputs; assess reliability and hallucination; apply basic safety/injection awareness in clinical apps.

Topics (theory) — 8 h:

#	Topic	h
8.1	Prompting fundamentals: zero/few-shot, roles, instructions	2
8.2	Chain-of-thought & reasoning prompts	1
8.3	Structured outputs (JSON, schemas), constrained generation	1
8.4	Retrieval-grounded prompting (intro to the RAG idea)	1
8.5	Reliability, hallucination & self-consistency; when NOT to trust	2
8.6	Safety & prompt-injection awareness in clinical apps	1

Datasets/tools: local LLM via Ollama (MedGemma) or HF; Python. **Assessment:** prompt-portfolio deliverable (50%); structured-extraction lab (30%); quiz (20%). **Key decisions:** prompting vs. fine-tuning vs. RAG; how to measure reliability; guardrails. **References:** plan §13 → *Medical LLMs*. **Hours:** Theory 8 + Lab 10 = 18.